

EMBEDDED SYSTEMS

Version 2021 - UTFPR

PROF. DOUGLAS RENAUX, PHD

BIG IDEAS
FOR EVERY SPACE

<https://www.renesas.com/br/en/img/misc/products-solutions/synergy-ssp-title.jpg>

2021 Prof. Douglas Renaux

SAMPLE MARKET SEGMENTS

- Consumer Electronics
- Telecommunications
- Home Automation
- Industrial Automation
- Transportation
 - Avionics
 - Navigation
 - Electric Vehicles
- Defense
- Medical Equipment
- ??? (many new areas to come)



Smart House



Smart Factory



Drone



Connected Care



Cool Gadget



Robots

source: Renesas DevCon2015

CONSUMER ELECTRONICS

- Phones
- Videogame consoles
- Printers
- Digital cameras
- Audio/Video:
 - Television
 - Music Players
 - Home Entertainment Systems
 - BRD players

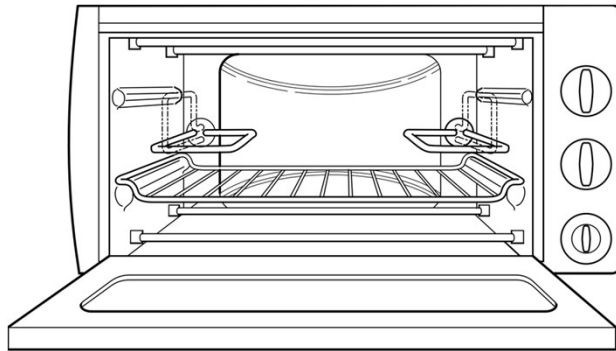


source: pixabay.com (CC)



HOUSEHOLD APPLIANCES

- Washing Machines
- Dishwasher
- Air Conditioners
- Microwave Oven



source: pixabay.com (CC)

TELECOMMUNICATIONS

- Routers
- Satellite Phones
- Switches



source: wikimedia.org (CC)



source: pixabay.com (CC)

HOME AUTOMATION

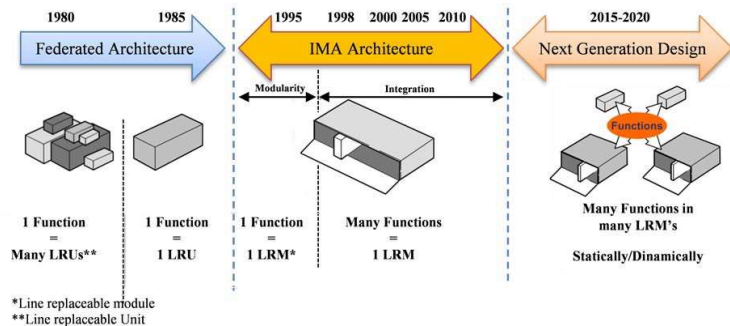


Smart Home



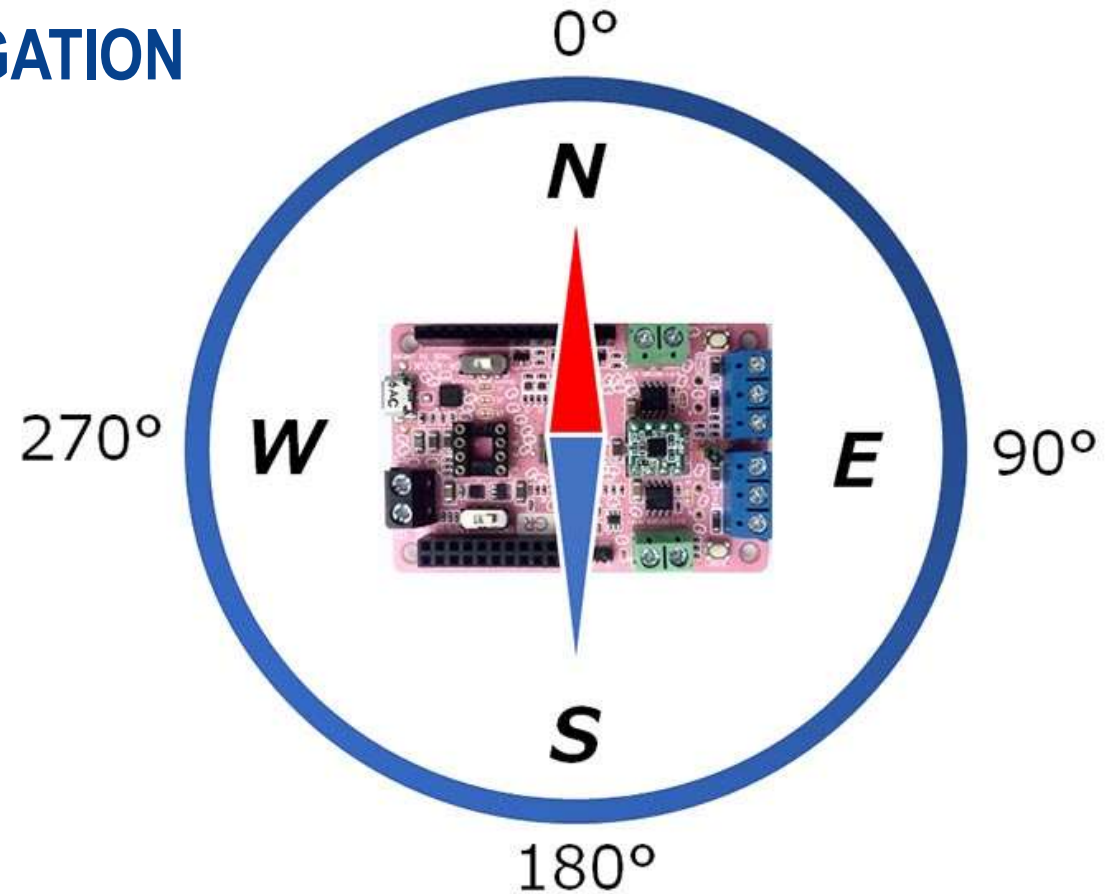
TRANSPORTATION - AVIONICS

Glass cockpit of the
Airbus A350 XWB



TRANSPORTATION - NAVIGATION

- Automotive GPS
- Electronic Compass

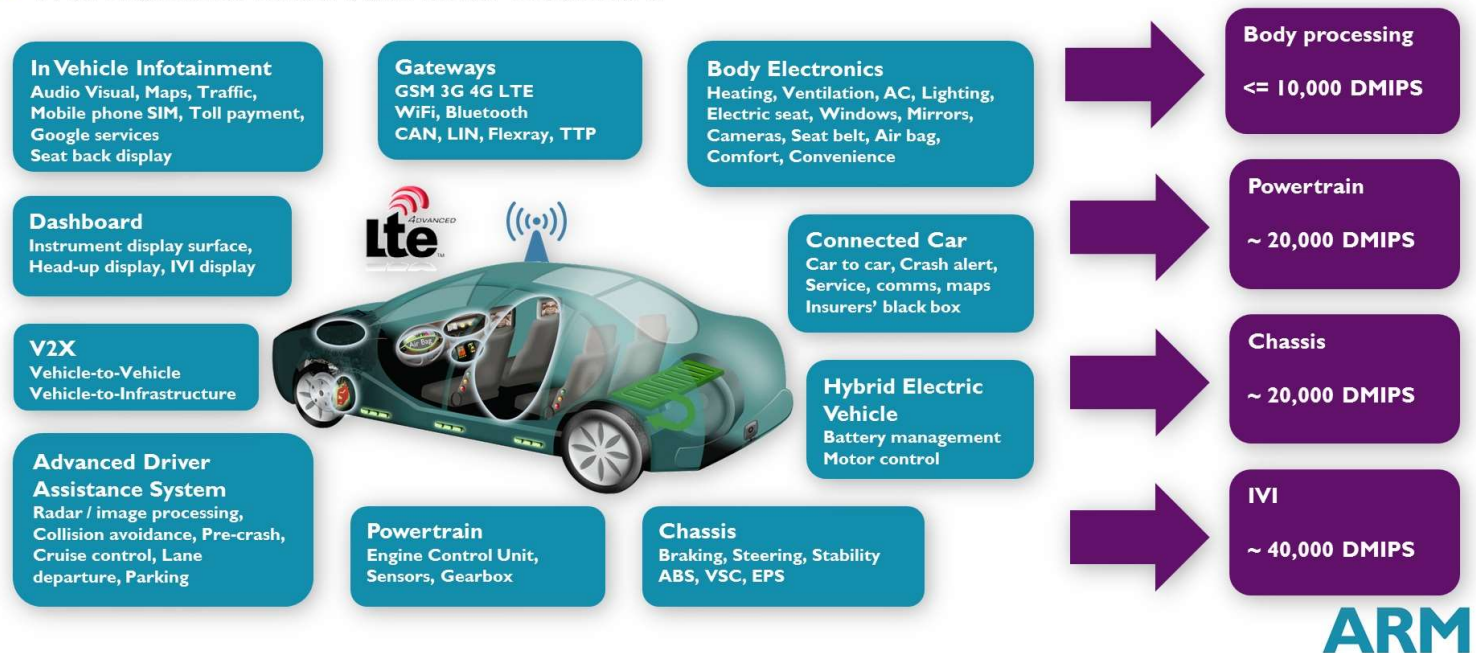


source: Renesas

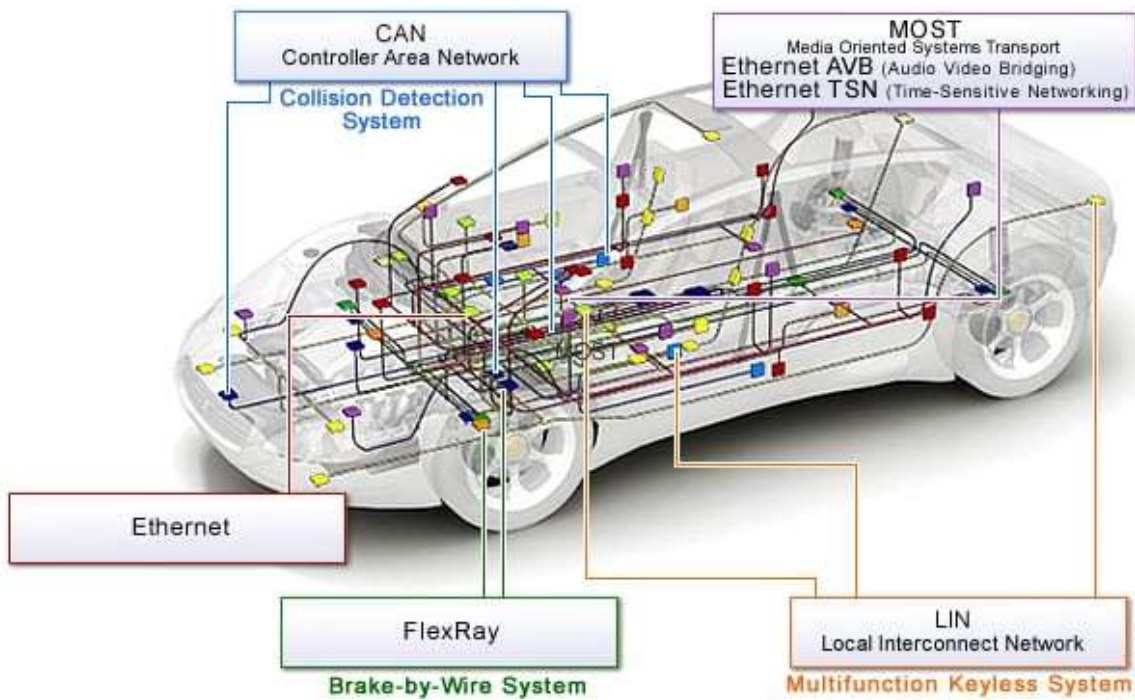
TRANSPORTATION - AUTOMOTIVE

Automotive ECUs Controllers by 2020

- Between 25 and 100 individual ECUs
- With distributed sensors and motor controllers.

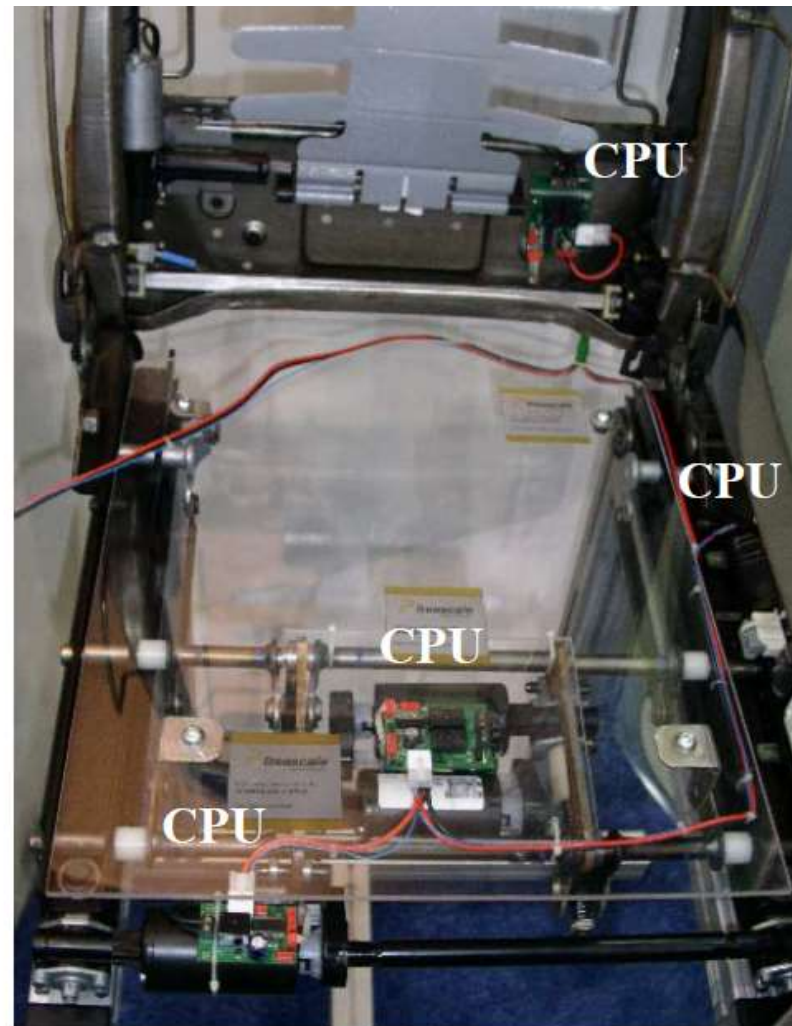


TRANSPORTATION - AUTOMOTIVE

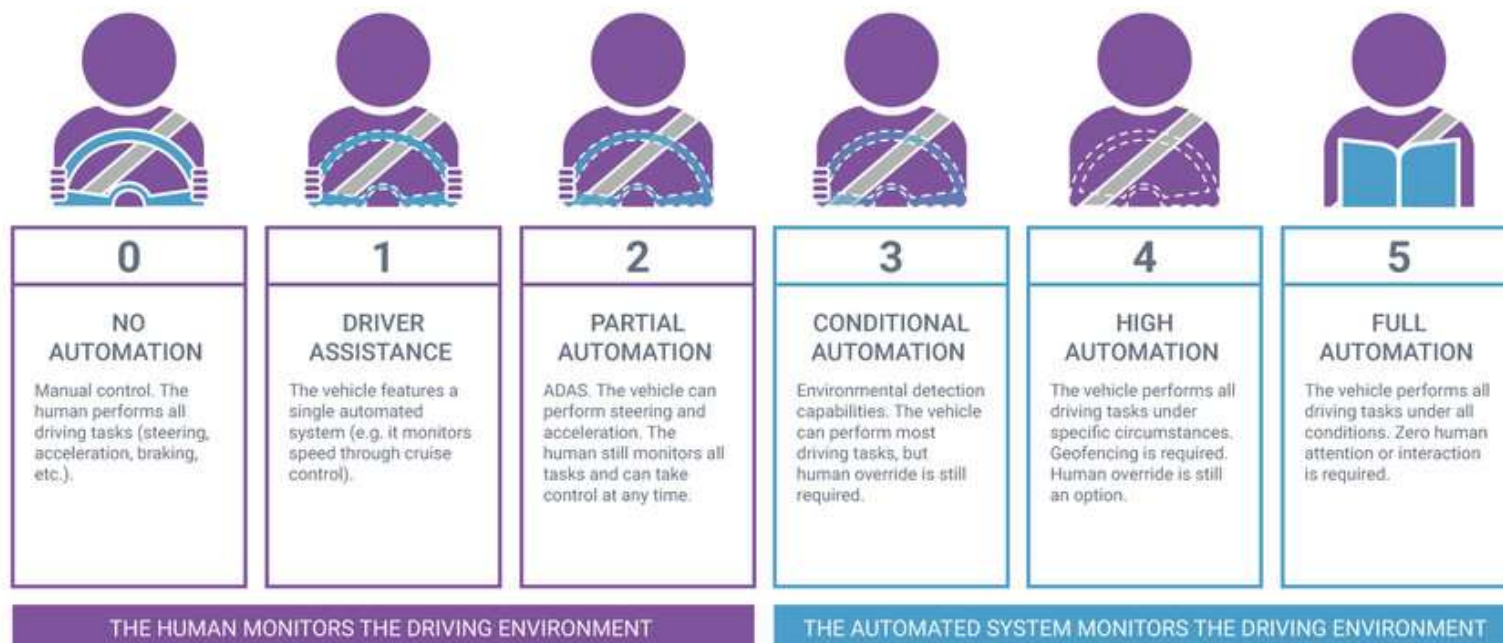


source: Renesas

- ◆ **Low speed LIN network to connect seat motion control nodes**
- ◆ **This is a distributed embedded system!**
 - Front-back motion
 - Seat tilt motion
 - Lumbar support
 - Control button interface



LEVELS OF DRIVING AUTOMATION

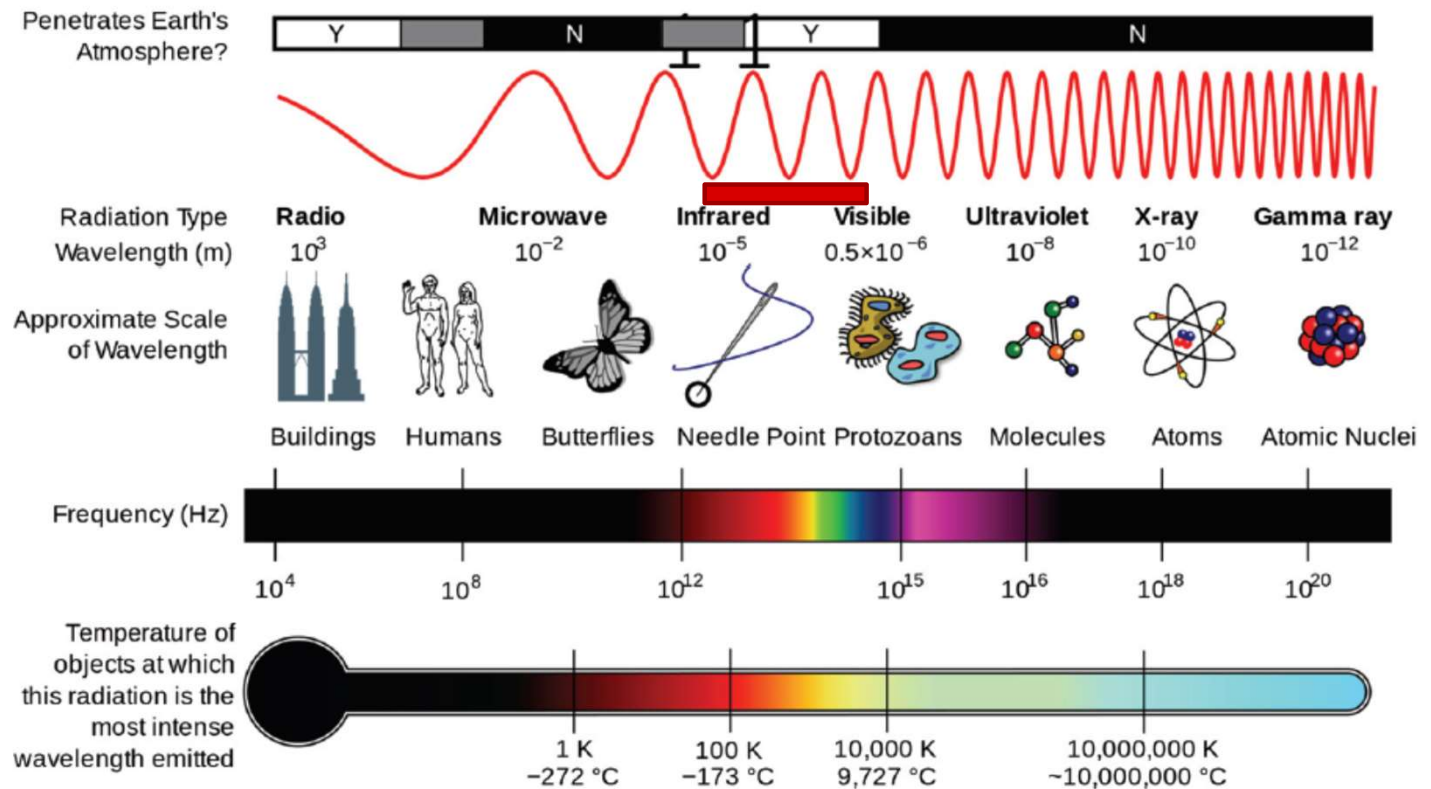


<https://www.synopsys.com/automotive/autonomous-driving-levels.html>

ADAS

ADVANCED DRIVER ASSISTANCE SYSTEM

Advance



MEDICAL EQUIPMENT

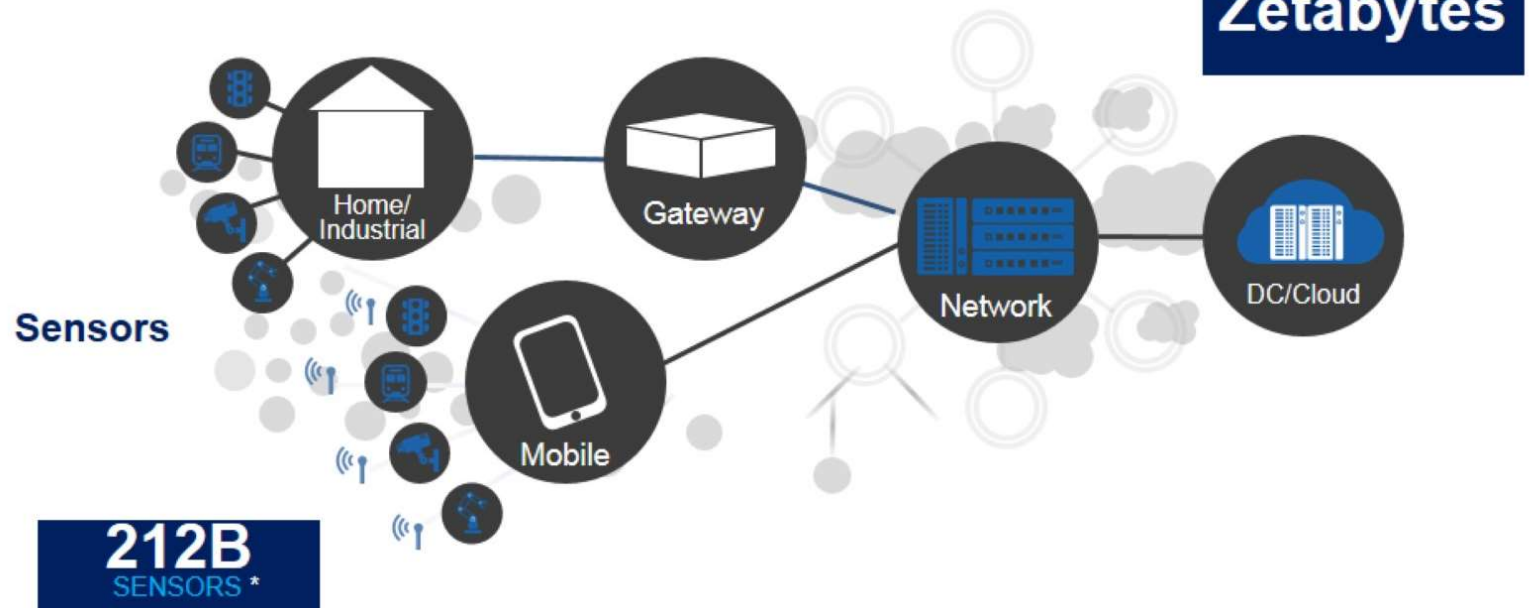
- CT Scanners
- ECG (Electrocardiogram)
- Blood Glucose Monitor
- Blood Pressure Monitor
- Body Composition Analyzer

**HEALTHCARE SOLUTION
USING RENESAS SYNERGY™**

Accelerate Medical Device Design with IEC62304 Class C Pre-Certified
Renesas Synergy™ Platform Safety Solution

THE IoT ERA

The Internet of Things is Everywhere



NOVOS SUB-SEGME

eVTOL - electrical Vertical

